```
clc;
clear;
close all;

% Experiment 4: FIR Filter Using Window Functions

Fs = 8000;
fc = 1500;
N = 50;

% Normalized Cutoff Frequency
Wn = fc / (Fs/2);

% Window Functions
w_rect = rectwin(N+1);
w_hamm = hamming(N+1);
w_hann = hann(N+1);
w_black = blackman(N+1);

% FIR Low-Pass Filter Design
b_rect = fir1(N, Wn, w_rect);
b_hamm = fir1(N, Wn, w_hamm);
b_hann = fir1(N, Wn, w_hann);
b_black = fir1(N, Wn, w_black);

% Frequency Response
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);

% Display Filter Coefficients
fprintf('FIR LOW-PASS FILTER USING WINDOW FUNCTIONS\n');
```

```
FIR LOW-PASS FILTER USING WINDOW FUNCTIONS
```

```
fprintf('Filter Order = %d\n', N);
```

```
Filter Order = 50
```

```
fprintf('Sampling Frequency = %d Hz\n', Fs);
```

```
Sampling Frequency = 8000 Hz
```

```
fprintf('Cutoff Frequency = %d Hz\n', fc);
```

```
Cutoff Frequency = 1500 Hz
```

```
fprintf('\nRectangular Window Coefficients (First 5):\n');
```

```
Rectangular Window Coefficients (First 5):
```

```
disp(b_rect(1:5));
```

```
   -0.0118         0    0.0128    0.0103   -0.0058
```

```
fprintf('Hamming Window Coefficients (First 5):\n');
```

```
Hamming Window Coefficients (First 5):
```

```
disp(b_hamm(1:5));
```

```
   -0.0009         0    0.0012    0.0011   -0.0008
```

```
fprintf('Hann Window Coefficients (First 5):\n');
```

```
Hann Window Coefficients (First 5):
```

```
disp(b_hann(1:5));
```

```
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588
```

```
fprintf('Blackman Window Coefficients (First 5):\n');
```

```
Blackman Window Coefficients (First 5):
```

```
disp(b_black(1:5));
```

```
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```

```
% Magnitude Response
figure('Name','FIR Window Comparison');

subplot(2,1,1);

plot(f, 20*log10(max(abs(H_rect),1e-5)), 'k', ...
    'LineWidth', 1.2);
hold on;

plot(f, 20*log10(max(abs(H_hamm),1e-5)), 'b', ...
    'LineWidth', 1.2);

plot(f, 20*log10(max(abs(H_hann),1e-5)), 'g', ...
    'LineWidth', 1.2);

plot(f, 20*log10(max(abs(H_black),1e-5)), 'r', ...
    'LineWidth', 1.2);
```

```
grid on;
legend('Rectangular','Hamming','Hann','Blackman');
title('FIR Low-Pass Filter Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([0 Fs/2]);
ylim([-100 10]);

% Phase Response
subplot(2,1,2);

plot(f, unwrap(angle(H_hamm))*180/pi, ...
    'b', 'LineWidth', 1.2);

grid on;
title('Phase Response of Hamming Window FIR Filter');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
xlim([0 Fs/2]);
```

